

□ DEFICIENCY & OTHER DISEASES — FULL MIND MAP

Biology Ch. 9.vi · Diseases & Treatment · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this sub-chapter:

1. Metabolic, Joint & Food-Adulteration Disorders
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4. Nuclear Medicine, Radioisotopes & Cancer
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6. Pollution, Heavy Metals & Occupational Diseases
7. Scurvy, Malnutrition & Deficiency Matching
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1. Metabolic, Joint & Food-Adulteration Disorders

- Muscle fatigue ko source anaerobic metabolism se banne wale lactic acid accumulation se link karta hai; local acidity muscular contraction ke liye unfavourable ho sakti hai. [Q1]
- Joints mein uric-acid crystal deposition arthritis/gout se linked hai; blood mein high uric acid (hyperuricemia) Gout ki characteristic hai. [Q2, Q3]
- Epidemic dropsy mustard oil ke Argemone-oil adulteration se associated hai. Source Q4 mein rice-husk adulteration ko dropsy ka “not correct” reason mark karta hai; Argemone mexicana poppy-family seed mustard seed jaisa dikh sakta hai. [Q4, Q5]
- Cancer = abnormal/uncontrolled cell growth with potential to invade/spread. Source ka quoted “single cell...merciless disease” answer Cancer hai. [Q6]
- Anthophobia = flowers ka fear. Source hysteria ko mainly young/unmarried women se associate karta hai. [Q7, Q8]
- Silicosis = silica particles inhale karne se hone wali lung disease. [Q9]

2. Iodine, Beriberi & Blue Baby Syndrome

- Iodine deficiency → thyroid malfunction / goitre; source thyroxine aur triiodothyronine production kam hone se link karta hai. Goitre patients ko iodine diya jaata hai. [Q10, Q12, Q13, Q14]
- Deficiency matching: Anaemia → Iron; Goitre → Iodine; Night-blindness → Vitamin A; Beriberi → Vitamin B (Thiamine/B1). [Q11]
- Polished rice ki milling/polishing nutrients remove karti hai; source ke according thiamine (Vitamin B1) deficiency ke kaaran Beriberi hota hai. [Q15, Q16, Q17]
- Blue Baby Syndrome: drinking water mein excess nitrate → nitrite → oxyhaemoglobin se methemoglobin; blood mein excess methemoglobin oxygen-carrying capacity ko reduce karta hai. [Q18, Q19, Q20]

3. Heart, Potassium & Kidney Disorders

- Heart-attack sequence source: cholesterol/fibrous plaque → blood clot → vessel lumen narrowing/obstruction → inadequate blood & oxygen supply. “Golden Hour” medical terminology heart attack se related hai. [Q21, Q22]
- Beta-blocker ko source heart attack se protection/management se link karta hai. Normal heart-beat regulation ke liye Potassium essential bataya gaya hai; potassium deficiency ka paired answer low blood pressure hai. [Q23, Q24, Q25]
- Urine ke saath albumin release hona kidney failure/damage ka indicator diya hai; prolonged fasting se source kidney ko most affected organ mark karta hai. [Q26, Q27]

4. Nuclear Medicine, Radioisotopes & Cancer

- Nuclear medicines = radioisotopes. Diagnostic matching: Arsenic-74 → tumour; Cobalt-60 → cancer; Iodine-131 → thyroid activity; Sodium-24 → blood disorder / blood-flow measurement. [Q28, Q29, Q31, Q36, Q37]
- Source Q30 ka incorrect match: Gold-198 → treatment of retinal disorders. Leukaemia mein WBC count abnormal increase hota hai. [Q30, Q32]
- Blood cancer/leukaemia control ke liye source Phosphorus-32 aur Cobalt-60 dono ko correct maanta hai; Cobalt-60 radiation treatment mein γ -rays emit karta hai. [Q33, Q34, Q35]
- Muscle/body ache relief ke liye Infrared radiation diya hai. [Q38, Q44]
- Oncogenes cancer se associated hain; cancer condition ko uncontrolled cell multiplication bataya gaya hai. Metastasis = cancer cells ka blood/lymph ke through other sites/organs tak spread. [Q39, Q40, Q41]
- Ionized radiation intensity/use ke depending cancer cause bhi kar sakti hai aur cure/treatment mein use bhi ho sakti hai. Harmful radiation ke liye source Brain ko listed organs mein less susceptible batata hai. [Q42, Q43]
- Source “most recent therapy” ka answer Intensity Modulated Radiotherapy (IMRT) deta hai; Interferon ko cancer control se link karta hai. [Q45, Q46]

EXAM TRAP — Radioisotope Answer Pattern

- ▶ Q33 mein source explicitly “Multiple Correct Answers — C & D” deta hai: Phosphorus-32 + Cobalt-60. Q34 mein isi wajah se “More than one of the above” correct hai. [Q33, Q34]

5. Brain, Nerve, Sleep & Behavioural Disorders

- Cerebral palsy ko source generally small children mein found brain disorder batata hai. Alzheimer’s primarily brain ko affect karta hai; source hazardous neural disease se Amyloid precursor protein ko associate karta hai. [Q47, Q48, Q49]
- Parkinson’s disease treatment development ke Nobel context mein source Arvid Carlsson ko answer karta hai. [Q50]
- Scratching itching ko ease karta hai—source certain nerves stimulation aur brain-mediated antihistaminic chemicals increase ka mechanism deta hai. [Q51]
- Anosmia = loss of smell. Matching mein Dyspnoea → shortness of breath aur Anosmia → partial/full loss of smell correct; Anorexia/Insomnia ke दिए गए swapped pairs incorrect hain. [Q52, Q53]
- Eating stop na kar paana = Bulimia. Sleep apnea = sleep ke dauran breathing briefly and repeatedly stop hona. [Q54, Q55]

6. Pollution, Heavy Metals & Occupational Diseases

- Disease–cause matching: Khaira → Zinc deficiency; Anaemia → Iron deficiency; Goitre → Iodine deficiency; Scurvy → Vitamin C deficiency. [Q56]
- Knock-knee syndrome ko source fluoride pollution se link karta hai; “Fluoride ion → Indigestion” ko incorrect pollutant–disease match mark karta hai. [Q57, Q58]
- Minamata disease = Mercury poisoning / mercury-polluted water / mercury-contaminated fish. Itai-Itai disease = chronic Cadmium poisoning/pollution. [Q59, Q60, Q61, Q62, Q63, Q64]
- Allergy-caused disease ka answer Asthma diya hai. [Q65]

7. Scurvy, Malnutrition & Deficiency Matching

- Milk + egg + bread-only diet se source Scurvy risk deta hai; Scurvy = Vitamin C deficiency, aur Gooseberry (Amla) treatment/useful source answer hai. [Q66, Q67, Q68, Q69]

- DRDO drug Lukoskin = Leucoderma treatment. Polyherbal product list mein source Nocdurna ko DRDO-developed nahi maanta. [Q70, Q71]
- Disease-affected part matching explanation: Malaria → blood cells; Filariasis → lymph node; Encephalitis → brain; Leukaemia → bone marrow. [Q72]
- Malnutrition matching: Marasmus → prolonged starvation; Kwashiorkor → protein deficiency; TB → bacterial infection; Hepatitis B → viral infection. [Q73]
- Kwashiorkor ko protein deficiency se directly link kiya hai. Source Q75 children mein protein deficiency ka answer Marasmus deta hai aur explanation prolonged starvation se energy + protein deficiency / generalized wasting batati hai. [Q74, Q75]

⚠ **EXAM TRAP — Marasmus vs Kwashiorkor**

- ▶ Q74: Kwashiorkor = protein deficiency. Q75: marked answer Marasmus; explanation ise prolonged starvation ke saath energy + protein deficiency se explain karti hai. Dono source framings ko separate yaad rakho. [Q74, Q75]
- ▶ Q72 ke printed code mein Malaria mapping “5” use hoti hai jabki visible List-II 1–4 tak hai; explanation nevertheless Malaria → Blood cells, Filariasis → Lymph node, Encephalitis → Brain, Leukaemia → Bone marrow deti hai. [Q72]

8. Disease–Organ Matching & Eye Disorders

- Correct disease–organ pairs: Jaundice → Liver; Pneumonia → Lungs. [Q76, Q77]
- Astigmatism aur Myopia dono eye disorders hain. Myopia/short-sightedness concave lens se correct hoti hai; myopic person nearby objects clearly dekh sakta hai. Hypermetropia/far-sightedness mein close objects dekhne mein difficulty hoti hai. [Q78, Q79, Q80, Q81, Q82]
- Blind spot aur yellow spot Retina mein hote hain; source Retina ko Diencephalon ka outgrowth batata hai. [Q83, Q84]
- Cataract = eye lens opaque/cloudy hona; listed iris/retina/cornea mein lens absent hone ki wajah se Q86 ka answer “None of the Above” hai. Cloudiness protein change se linked hai. [Q85, Q86, Q87]
- Drug/use matching: Atropine → pupil dilation; Ether → local anaesthesia; Nitroglycerine → heart trouble; Pyrethrin → mosquito control. Dilute atropine eye pupil dilate karta hai. [Q88, Q89]
- Ishihara test = colour blindness. [Q90]
- Biological resurfacing ko source knee/ankle/hip/vertebral joints ke context mein discuss karta hai; initial key knee-joint thi, revised key mein question delete hua. [Q91]

⚠ **EXAM TRAP — Deleted / Disputed Question**

- ▶ Q91 source mein “Disputed Question” hai: initial UPPSC key option (A) Knee joints thi, later revised key mein question deleted bataya gaya hai. [Q91]

9. Anaemia, Liver, Prions, Parasites & Other Disorders

- Iron deficiency → Anaemia. [Q92, Q93]
- Excessive alcoholic beverage → Liver cirrhosis. Oral submucous fibrosis → Gutka/tobacco consumption. Jaundice liver ko affect karta hai aur source use “not so contagious” disease ke answer ke roop mein deta hai. [Q94, Q95, Q96, Q97]
- Poultry-bird rickets = Vitamin D deficiency. Mad-cow disease factor = Prions. Liver fluke bile duct mein source host Sheep deta hai. [Q98, Q99, Q100]
- Bright’s disease Kidney ko affect karti hai. Pneumoconiosis mainly coal-mining workers se linked hai. [Q101, Q102]

10. Diabetes Mellitus / Insipidus & Vasopressin

- Blood glucose commonly mg/dL (milligram per decilitre) mein express hota hai. Excess insulin injection se glucose unusually low (hypoglycemia) ho sakta hai, source spasm/apoplexy/faintness/death se link karta hai. [Q103, Q104]
- Diabetes vaccination se control nahi hoti. Source diabetes ko Pancreas malfunction aur insulin deficiency se link karta hai. [Q105, Q106, Q107, Q108]
- Diabetes mellitus association: higher blood sugar + lower insulin. HbA1c management mein previous ~8–12 weeks ka average glucose reflect karta hai. [Q109, Q110]
- Diabetes insipidus: excessive urine + thirst; source ke according ADH/vasopressin production deficiency se hota hai, excess secretion se nahi. [Q111]
- Alcohol/whiskey urination increase karta hai because source vasopressin (ADH) secretion suppress hone aur diuretic effect ko describe karta hai. [Q112]
- Fenugreek seeds aur Curry leaf ko source diabetes blood-sugar control/medication benefit se associate karta hai. [Q113, Q114]
- Insulin discovery context: source F.G. Banting ko answer karta hai; explanation Banting & Best, Macleod aur 1923 Nobel context deti hai. [Q115]

11. Treatment, Emergency Medicine & Miscellaneous

- Copper-T ka commonest side effect source mein increased menstrual bleeding hai. [Q116]
- Emphysema environmental-pollution context mein lungs ko affect karta hai. [Q117]
- Aspirin = non-narcotic analgesic; “belongs to narcotic analgesics” statement source mein NOT true hai. Low-dose aspirin heart attack/stroke/blood-clot prevention context mein diya hai. [Q118]
- CPR = Cardiopulmonary Resuscitation. Coronary bypass graft ke liye source leg vein / chest or wrist artery jaise healthy vessels use hone ke kaaran “Any of these” answer deta hai. Cath lab = Cardiology department. [Q119, Q120, Q121]
- Xerophthalmia = severe Vitamin A deficiency; source India mein remedial priority aur preventable childhood blindness context deta hai. [Q122]
- Lathyrism = Khesari dal (Lathyrus sativus) ka excessive/regular consumption; neurological weakness, spasms aur lower-limb paralysis se linked. [Q123, Q124]
- Blank dispensing capsules mainly Gelatine se bante hain, lekin source Starch capsules ko bhi valid alternative maanta hai. [Q125]

EXAM TRAP — Multiple-Correct Capsule Question

- ▶ Q125 source explicitly “Multiple Correct Answers — C & D” deta hai: Starch + Gelatine. [Q125]